

# Comments

## Comments

- **Definition:** A comment that spans a single line.
- **Syntax:** Starts with `//` and extends to the end of the line.

javascript

Copy code

```
// This is a single-line comment
```

```
let x = 42; // This is a comment after the code
```

- **Use Cases:**
  - For short explanations or notes.
  - Temporarily disabling code for testing.

## Multi-line (Block) Comments

- **Definition:** A comment that spans multiple lines.
- **Syntax:** Begins with `/*` and ends with `*/`.

javascript

Copy code

```
/*
```

```
This is a block comment
```

```
It can span multiple lines
```

```
*/
```

```
let x = 42;
```

- **Use Cases:**
  - For explaining large blocks of code.
  - Commenting out larger sections of code for testing or debugging.

## Why Use Comments?

- **Purpose:**
  - To improve code readability and understandability.
  - Helps explain complex logic.

- Makes code easier to maintain.
- Helps with debugging by explaining code intentions or expected behavior.
- **Best Practices:**
  - Use comments when the code is complex or non-obvious.
  - Avoid over-commenting; code should be self-explanatory when possible.

#### **Good Example:**

javascript

Copy code

```
angle = angle + 90; // Rotate 90 degrees to compensate for vertical screen
```

#### **Bad Example:**

javascript

Copy code

```
// setting greetingText to Hello
```

```
const greetingText = "Hello";
```

#### **Documentation with Comments**

- **Definition:** Use comments to describe functions, parameters, and the logic behind the code.
- **Tools:** Documentation tools (e.g., JSDoc) can parse these comments and generate formal documentation.

javascript

Copy code

```
/**  
 * Function to calculate the area of a circle  
 * @param {number} radius - The radius of the circle  
 * @returns {number} - The area of the circle  
 */  
function calculateArea(radius) {  
    return Math.PI * Math.pow(radius, 2);  
}
```

#### **Code Toggle (Commenting Out Code)**

- **Definition:** Temporarily disabling parts of the code for testing or debugging.

javascript

Copy code

```
// const greetingText = "Hi";  
const greetingText = "Hello"; // Now active  
// const greetingText = "Welcome"; // Disabled for testing
```

- **Use Cases:**
  - Troubleshooting a problematic code section.
  - Testing alternate versions of code.
  - Experimenting with changes while retaining original code.

### Common Commenting Shortcuts in Editors

- **Windows:** Ctrl + /
- **macOS:** Cmd + /

These shortcuts allow you to quickly comment or uncomment selected lines of code.

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### Quiz Questions

1. **What is the syntax for a single-line comment in JavaScript?**
  - Answer: // comment here
2. **What is the syntax for a multi-line (block) comment in JavaScript?**
  - Answer: /\* comment here \*/
3. **What are comments used for in JavaScript?**
  - Answer: Comments are used to explain code, improve readability, document functions, and temporarily disable parts of code.
4. **What is the proper use of a single-line comment?**
  - Answer: Single-line comments are used for brief explanations or notes next to a line of code.
5. **What is the purpose of multi-line comments?**
  - Answer: Multi-line comments are used for longer explanations or to comment out larger sections of code.
6. **Why should comments be used sparingly in code?**

- Answer: Over-commenting can make the code harder to read. It's better to write self-explanatory code and comment when necessary.

**7. Which tool can be used to generate documentation from comments in JavaScript?**

- Answer: JSDoc

**8. What are the keyboard shortcuts to comment code in Visual Studio Code (VS Code) on Windows and macOS?**

- Answer: Ctrl + / for Windows, Cmd + / for macOS.

**9. What does the following code do?**

javascript

Copy code

```
// const greetingText = "Hi";
```

```
const greetingText = "Hello";
```

- Answer: The first line is commented out, so "Hello" is assigned to the greetingText variable.

**10. How can comments help during the debugging process?**

- Answer: Comments can temporarily disable code, making it easier to identify issues by testing different parts of the code without deleting them.